A RING IS JACOBSON EXACTLY WHEN ITS SPECTRUM IS A JACOBSON TOPOLOGICAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof refer to a definition of the Topology chapter and to a same-chapter lemma; both are recorded as navigation stubs pointed at their \label lines, the cross-chapter one carrying kind definition.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- The topological notion of a Jacobson space. Invoked in the statement and used at the start of the counted proof (lines 6910-6919). Supplies the exact condition on closed subsets of the spectrum that has to be matched.
- Zariski closed subsets correspond to radical ideals. Invoked at the end of the counted proof (lines 6939-6941). Converts the equality of two closed subsets into equality of the corresponding radical ideals.

Source reference. Original lemma 00E0: sources/stacks-algebra/pinned-original-source.tex, source lines 3204–3204. Navigation pointer for \ref{lemma-Zariski-topology} used inside the counted span (line 6924); the argument that consumes it is retained in the span.

Source reference. Original definition 005U: sources/stacks-chow/tags.txt, source lines 222–222. Navigation pointer for \ref{topology-definition-space-jacobson} used inside the counted span (line 6912); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *A ring R is Jacobson if and only if* $\mathrm{Spec}(R)$ *is Jacobson, see Topology, Definition 005U.*

Proof. Suppose R is Jacobson. Let $Z \subset \mathrm{Spec}(R)$ be a closed subset. We have to show that the set of closed points in Z is dense in Z. Let $U \subset \mathrm{Spec}(R)$ be an open such that $U \cap Z$ is nonempty. We have to show $Z \cap U$ contains a closed point of $\mathrm{Spec}(R)$. We may assume $U = D(f)$ as standard opens form a basis for the topology on $\mathrm{Spec}(R)$. According to Lemma 00E0 we may assume that $Z = V(I)$, where I is a radical ideal. We see also that $f \notin I$. By assumption, there exists a maximal ideal $\mathfrak{m} \subset R$ such that $I \subset \mathfrak{m}$ but $f \notin \mathfrak{m}$. Hence $\mathfrak{m} \in D(f) \cap V(I) = U \cap Z$ as desired.

Conversely, suppose that $\mathrm{Spec}(R)$ is Jacobson. Let $I \subset R$ be a radical ideal. Let $J = \cap_{I \subset \mathfrak{m}} \mathfrak{m}$ be the intersection of the maximal ideals containing I. Clearly J is a radical ideal, $V(J) \subset V(I)$, and $V(J)$ is the smallest closed subset of $V(I)$ containing all the closed points of $V(I)$. By assumption we see that $V(J) = V(I)$. But Lemma 00E0 shows there is a bijection between Zariski closed sets and radical ideals, hence $I = J$ as desired. □